

Extracting Data from Multiple Sources



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Overview



Azure Databricks support connectors for various data sources

Mount Azure Storage & Data Lake Store

Service Principal authentication

Read multiple file formats like CSV, TSV & JSON

Use options for extraction

Apply schema to JSON documents



Extracting from Azure Storage Services



Data Sources



Relational databases using JDBC

NoSQL connectors

- MongoDB, Neo4j, Cassandra etc.

Azure data connectors

- Azure SQL, SQL DW, Cosmos DB etc.

Mount Azure storage services

- Azure Storage, Data Lake Store (Gen1 & Gen2)

Mounting storage account to DBFS
allows seamless access to data without
requiring credentials



Mounting Storage to DBFS



No need to provide credentials every time

Access storage using file system semantics instead of URLs

Files persist to storage

Mount Azure Blob Storage using Access Key or SAS

Mount Azure Data Lake (Gen1 & Gen2) using Service Principal

Azure Service Principal



Identity created for use with applications to access Azure resources



With applications, use Service Principal instead of a user's Azure Active Directory (AD) credentials



To use it, register an application in Azure AD and create a secret



Use App ID and Secret to access Azure resources



Databricks Utilities



Set of utilities to perform powerful tasks inside notebooks

Various types

- File system, Notebook, Library utilities etc.

Example

- Mount a storage account
- Invoke one notebook from another
- Install a library on cluster from notebook

Reading Multiple File Formats



Applying Schemas



Summary



Support extraction from relational, non-relational and file based storage

Mounted Azure Storage and Data Lake to DBFS using different authentication ways

Use Databricks Utilities to explore data

Read data from CSV, TSV and JSON files

Used options

- Delimiter, Header, Multiline etc.

Built and applied schema to files

